

hw0428

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1 Exercise 10.18: Mroz data

```
mroz <- read.csv("mroz.csv")

# The question asks for the 428 married women who participate in the labor force.
mroz_work <- subset(mroz, lfp == 1 & wage > 0)

mroz_work$lwage <- log(mroz_work$wage)
mroz_work$exper2 <- mroz_work$exper^2
mroz_work$MOTHERCOLL <- ifelse(mroz_work$mothereduc > 12, 1, 0)
mroz_work$FATHERCOLL <- ifelse(mroz_work$fathereduc > 12, 1, 0)

nrow(mroz_work)
```

```
## [1] 428
```

1.1 10.18(a)

```
parent_percent <- data.frame(
  Variable = c("MOTHERCOLL", "FATHERCOLL", "All parents combined"),
  Percent = c(
    mean(mroz_work$MOTHERCOLL) * 100,
    mean(mroz_work$FATHERCOLL) * 100,
    mean(c(mroz_work$MOTHERCOLL, mroz_work$FATHERCOLL)) * 100
  )
)

kable_num(parent_percent, caption = "Percentage of parents with some college education", row.names = FALSE)
```

Percentage of parents with some college education

Variable	Percent
MOTHERCOLL	12.149
FATHERCOLL	11.682
All parents combined	11.916

Answer. About 12.15% of mothers and 11.68% of fathers have some college education. Combining mothers and fathers, the parent-level percentage is about 11.92%.

1.2 10.18(b)

```
corr_mat <- cor(mroz_work[, c("educ", "MOTHERCOLL", "FATHERCOLL")])
kable_num(corr_mat, caption = "Correlations among EDUC, MOTHERCOLL, and FATHERCOLL")
```

Correlations among EDUC, MOTHERCOLL, and FATHERCOLL

	educ	MOTHERCOLL	FATHERCOLL
educ	1.0000	0.3595	0.3985
MOTHERCOLL	0.3595	1.0000	0.3546
FATHERCOLL	0.3985	0.3546	1.0000

Answer. The correlations between EDUC and the two proposed instruments are moderate: MOTHERCOLL is correlated with EDUC by about 0.359, and FATHERCOLL is correlated with EDUC by about 0.398. These magnitudes matter because instruments must be relevant. The college dummies may be better than raw parent education because the college threshold may capture a stronger non-linear family education effect, while using fewer direct details about parental background.

1.3 10.18(c)

The wage equation from Example 10.5 is taken to be

$$\log(WAGE) = \beta_0 + \beta_1 EDUC + \beta_2 EXPER + \beta_3 EXPER^2 + u.$$

Here, EDUC is treated as endogenous and MOTHERCOLL is used as the excluded instrument.

```
y_mroz <- mroz_work$lwage
X_mroz <- model.matrix(~ educ + exper + exper2, data = mroz_work)
Z_mother <- model.matrix(~ exper + exper2 + MOTHERCOLL, data = mroz_work)

iv_mother <- iv_2sls(y_mroz, X_mroz, Z_mother)
kable_num(iv_mother$table, caption = "2SLS using MOTHERCOLL as the instrument", row.names = FALSE)
```

2SLS using MOTHERCOLL as the instrument

term	estimate	std_error	t_value	p_value	conf_low	conf_high
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term	estimate	std_error	t_value	p_value	conf_low	conf_high
(Intercept)	-0.1328	0.4965	-0.2674	0.7893	-1.1087	0.8432
educ	0.0760	0.0394	1.9290	0.0544	-0.0014	0.1535
exper	0.0433	0.0134	3.2314	0.0013	0.0170	0.0697
exper2	-0.0009	0.0004	-2.1687	0.0307	-0.0017	-0.0001

```
educ_ci_mother <- subset(iv_mother$stable, term == "educ")[, c("estimate", "conf_low",
"conf_high")]
kable_num(educ_ci_mother, caption = "95% CI for the coefficient of EDUC using MOTHERCOLL", row.names = FALSE)
```

95% CI for the coefficient of EDUC using MOTHERCOLL

estimate	conf_low	conf_high
0.076	-0.0014	0.1535

Answer. Using MOTHERCOLL as the instrument, the IV estimate of the return to education is about 0.076. The 95% confidence interval is approximately [-0.0014, 0.1535].

1.4 10.18(d)

```
fs_mother <- lm(educ ~ exper + exper2 + MOTHERCOLL, data = mroz_work)
fs_mother_restricted <- lm(educ ~ exper + exper2, data = mroz_work)

kable_num(coef(summary(fs_mother)), caption = "First-stage regression using MOTHERCOLL")
```

First-stage regression using MOTHERCOLL

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.0791	0.3031	39.8494	0.0000
exper	0.0562	0.0421	1.3356	0.1824
exper2	-0.0020	0.0013	-1.5574	0.1201
MOTHERCOLL	2.5171	0.3157	7.9726	0.0000

```
F_mother <- joint_F(fs_mother_restricted, fs_mother, q = 1)
kable_num(F_mother, caption = "First-stage F test for MOTHERCOLL")
```

First-stage F test for MOTHERCOLL

F	df1	df2	p_value
63.563	1	424	0

Answer. The first-stage F statistic for MOTHERCOLL is about 63.56. Since this is much larger than the common rule-of-thumb value 10, MOTHERCOLL appears to be a strong instrument.

1.5 10.18(e)

Now use both MOTHERCOLL and FATHERCOLL as excluded instruments.

```
Z_both_parents <- model.matrix(~ exper + exper2 + MOTHERCOLL + FATHERCOLL, data = mro
z_work)
iv_both_parents <- iv_2sls(y_mroz, X_mroz, Z_both_parents)

kable_num(iv_both_parents$table, caption = "2SLS using MOTHERCOLL and FATHERCOLL as i
nstruments", row.names = FALSE)
```

2SLS using MOTHERCOLL and FATHERCOLL as instruments

term	estimate	std_error	t_value	p_value	conf_low	conf_high
(Intercept)	-0.2791	0.3922	-0.7115	0.4771	-1.0500	0.4919
educ	0.0878	0.0308	2.8540	0.0045	0.0273	0.1483
exper	0.0427	0.0133	3.2099	0.0014	0.0165	0.0688
exper2	-0.0008	0.0004	-2.1345	0.0334	-0.0016	-0.0001

```
educ_ci_both <- subset(iv_both_parents$table, term == "educ")[, c("estimate", "conf_l
ow", "conf_high")]
kable_num(educ_ci_both, caption = "95% CI for the coefficient of EDUC using both inst
ruments", row.names = FALSE)
```

95% CI for the coefficient of EDUC using both instruments

estimate	conf_low	conf_high
0.0878	0.0273	0.1483

```
ci_compare_mroz <- data.frame(
  Method = c("MOTHERCOLL only", "MOTHERCOLL + FATHERCOLL"),
  Estimate = c(educ_ci_mother$estimate, educ_ci_both$estimate),
  Lower = c(educ_ci_mother$conf_low, educ_ci_both$conf_low),
  Upper = c(educ_ci_mother$conf_high, educ_ci_both$conf_high),
  Width = c(educ_ci_mother$conf_high - educ_ci_mother$conf_low,
            educ_ci_both$conf_high - educ_ci_both$conf_low)
)

kable_num(ci_compare_mroz, caption = "Comparison of EDUC confidence intervals", row.n
ames = FALSE)
```

Comparison of EDUC confidence intervals

Method	Estimate	Lower	Upper	Width
MOTHERCOLL only	0.0760	-0.0014	0.1535	0.1549

Method	Estimate	Lower	Upper	Width
MOTHERCOLL + FATHERCOLL	0.0878	0.0273	0.1483	0.1210

Answer. With both instruments, the 95% confidence interval for EDUC is approximately [0.0273, 0.1483]. This interval is narrower than the interval in part (c).

1.6 10.18(f)

```
fs_both_parents <- lm(educ ~ exper + exper2 + MOTHERCOLL + FATHERCOLL, data = mroz_work)
fs_both_restricted <- lm(educ ~ exper + exper2, data = mroz_work)

kable_num(coef(summary(fs_both_parents)), caption = "First-stage regression using both parent college instruments")
```

First-stage regression using both parent college instruments

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.8903	0.2903	40.9655	0.0000
exper	0.0491	0.0401	1.2246	0.2214
exper2	-0.0014	0.0012	-1.2086	0.2275
MOTHERCOLL	1.7499	0.3223	5.4288	0.0000
FATHERCOLL	2.1866	0.3299	6.6278	0.0000

```
F_both_parents <- joint_F(fs_both_restricted, fs_both_parents, q = 2)
kable_num(F_both_parents, caption = "Joint first-stage F test for MOTHERCOLL and FATHERCOLL")
```

Joint first-stage F test for MOTHERCOLL and FATHERCOLL

F	df1	df2	p_value
56.963	2	423	0

Answer. The joint first-stage F statistic is about 56.96. Therefore, the two instruments seem adequately strong.

1.7 10.18(g)

Since there are two excluded instruments for one endogenous regressor, there is one surplus instrument. We use a Sargan overidentification test.

```
Z_mroz_sargan <- data.frame(
  exper = mroz_work$exper,
  exper2 = mroz_work$exper2,
  MOTHERCOLL = mroz_work$MOTHERCOLL,
  FATHERCOLL = mroz_work$FATHERCOLL
)

sargan_mroz <- sargan_test(iv_both_parents$residuals, Z_mroz_sargan, df_overid = 1)
kable_num(sargan_mroz, caption = "Sargan test for the surplus instrument")
```

Sargan test for the surplus instrument

	statistic	df	p_value
	0.2376	1	0.626

Answer. The Sargan p-value is about 0.626. At the 5% level, we fail to reject the validity of the overidentifying restriction. This supports the surplus instrument, although it does not prove instrument exogeneity.

2 Exercise 10.20: CAPM data

```
capm5 <- read.csv("capm5.csv")

capm5$msftrf <- capm5$msft - capm5$riskfree
capm5$mktrf <- capm5$mkt - capm5$riskfree

nrow(capm5)
```

```
## [1] 180
```

2.1 10.20(a)

```
ols_capm <- lm(msftrf ~ mktrf, data = capm5)
kable_num(coef(summary(ols_capm)), caption = "OLS CAPM estimate for Microsoft")
```

OLS CAPM estimate for Microsoft

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0032	0.0060	0.5384	0.591
mktrf	1.2018	0.1222	9.8389	0.000

Answer. The OLS beta estimate is about 1.2018. Since it is greater than one, Microsoft appears riskier than the market portfolio during this period.

2.2 10.20(b)

```
capm5$RANK <- rank(capm5$mktrf, ties.method = "first")

fs_rank <- lm(mktrf ~ RANK, data = capm5)
kable_num(coef(summary(fs_rank)), caption = "First-stage regression using RANK")
```

First-stage regression using RANK

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0790	0.0022	-35.998	0
RANK	0.0009	0.0000	43.100	0

```
fs_rank_summary <- data.frame(
  R_squared = summary(fs_rank)$r.squared,
  F_statistic = unname(summary(fs_rank)$fstatistic[1]),
  p_value = pf(summary(fs_rank)$fstatistic[1],
               summary(fs_rank)$fstatistic[2],
               summary(fs_rank)$fstatistic[3],
               lower.tail = FALSE)
)
kable_num(fs_rank_summary, caption = "First-stage strength using RANK", row.names = FALSE)
```

First-stage strength using RANK

	R_squared	F_statistic	p_value
	0.9126	1857.6	0

Answer. RANK is extremely significant in the first-stage regression, and the first-stage R^2 is about 0.9126. It is a strong predictor of the observed market risk premium. However, because RANK is constructed directly from the same measured market return, its exogeneity is questionable, so it may not satisfy all IV conditions.

2.3 10.20(c)

```
capm5$vhat_rank <- resid(fs_rank)
aug_rank <- lm(msftrf ~ mktrf + vhat_rank, data = capm5)
kable_num(coef(summary(aug_rank)), caption = "Augmented CAPM equation using first-stage residual from RANK")
```

Augmented CAPM equation using first-stage residual from RANK

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0030	0.0060	0.5044	0.6146
mktrf	1.2783	0.1267	10.0854	0.0000
vhat_rank	-0.8746	0.4286	-2.0405	0.0428

```
p_vhat_rank <- coef(summary(aug_rank))["vhat_rank", "Pr(>|t|)"]
```

Answer. The p-value for the first-stage residual is about 0.0428. At the 1% level, we fail to reject exogeneity of the market risk premium. This is not proof of exogeneity, but there is not enough evidence against it at 1%.

2.4 10.20(d)

```
y_capm <- capm5$msftrf
X_capm <- model.matrix(~ mktrf, data = capm5)
Z_rank <- model.matrix(~ RANK, data = capm5)

iv_rank <- iv_2sls(y_capm, X_capm, Z_rank)
kable_num(iv_rank$table, caption = "2SLS CAPM estimate using RANK as IV", row.names = FALSE)
```

2SLS CAPM estimate using RANK as IV

term	estimate	std_error	t_value	p_value	conf_low	conf_high
(Intercept)	0.0030	0.006	0.4994	0.6181	-0.0089	0.0149
mktrf	1.2783	0.128	9.9860	0.0000	1.0257	1.5309

```
compare_rank_ols <- data.frame(
  Method = c("OLS", "IV with RANK"),
  Beta = c(coef(ols_capm)["mktrf"], subset(iv_rank$table, term == "mktrf")$estimate)
)
kable_num(compare_rank_ols, caption = "Comparison of OLS and IV beta estimates", row.names = FALSE)
```

Comparison of OLS and IV beta estimates

Method	Beta
OLS	1.2018
IV with RANK	1.2783

Answer. The IV beta using RANK is about 1.2783, which is larger than the OLS beta. This agrees with the usual measurement-error expectation that OLS may be attenuated toward zero.

2.5 10.20(e)

```
capm5$POS <- ifelse(capm5$mktrf > 0, 1, 0)

fs_rank_pos <- lm(mktrf ~ RANK + POS, data = capm5)
fs_rank_pos_restricted <- lm(mktrf ~ 1, data = capm5)

kable_num(coef(summary(fs_rank_pos)), caption = "First-stage regression using RANK and POS")
```


First-stage regression using RANK and POS

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0804	0.0023	-35.5506	0.0000
RANK	0.0010	0.0000	24.5487	0.0000
POS	-0.0093	0.0042	-2.2005	0.0291

```
F_rank_pos <- joint_F(fs_rank_pos_restricted, fs_rank_pos, q = 2)
kable_num(F_rank_pos, caption = "Joint first-stage F test for RANK and POS", row.names = FALSE)
```

Joint first-stage F test for RANK and POS

F	df1	df2	p_value
951.26	2	177	0

```
fs_rank_pos_summary <- data.frame(
  R_squared = summary(fs_rank_pos)$r.squared
)
kable_num(fs_rank_pos_summary, caption = "First-stage R-squared using RANK and POS",
row.names = FALSE)
```

First-stage R-squared using RANK and POS

R_squared
0.9149

Answer. The joint first-stage F statistic is about 951.26, and the first-stage R^2 is about 0.9149. These instruments are jointly very strong in the first stage, although validity is still a separate issue.

2.6 10.20(f)

```
capm5$vhat_rank_pos <- resid(fs_rank_pos)
aug_rank_pos <- lm(msftrf ~ mktrf + vhat_rank_pos, data = capm5)
kable_num(coef(summary(aug_rank_pos)), caption = "Augmented CAPM equation using first
-stage residual from RANK and POS")
```

Augmented CAPM equation using first-stage residual from RANK and POS

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0030	0.0060	0.5029	0.6157
mktrf	1.2831	0.1263	10.1557	0.0000
vhat_rank_pos	-0.9549	0.4331	-2.2050	0.0287

```
p_vhat_rank_pos <- coef(summary(aug_rank_pos))["vhat_rank_pos", "Pr(>|t|)"]
```

Answer. The p-value for the first-stage residual is about 0.0287. At the 1% level, we fail to reject exogeneity of the market risk premium.

2.7 10.20(g)

```
Z_rank_pos <- model.matrix(~ RANK + POS, data = capm5)
iv_rank_pos <- iv_2sls(y_capm, X_capm, Z_rank_pos)
kable_num(iv_rank_pos$table, caption = "2SLS CAPM estimate using RANK and POS as IVs", row.names = FALSE)
```

2SLS CAPM estimate using RANK and POS as IVs

term	estimate	std_error	t_value	p_value	conf_low	conf_high
(Intercept)	0.0030	0.0060	0.4969	0.6199	-0.0089	0.0149
mktrf	1.2831	0.1279	10.0349	0.0000	1.0308	1.5354

```
compare_rank_pos_ols <- data.frame(
  Method = c("OLS", "IV with RANK and POS"),
  Beta = c(coef(ols_capm)["mktrf"], subset(iv_rank_pos$table, term == "mktrf")$estimate)
)
kable_num(compare_rank_pos_ols, caption = "Comparison of OLS and IV beta estimates", row.names = FALSE)
```

Comparison of OLS and IV beta estimates

Method	Beta
OLS	1.2018
IV with RANK and POS	1.2831

Answer. The IV beta using RANK and POS is about 1.2831, again larger than the OLS beta. This is consistent with the measurement-error story.

2.8 10.20(h)

For the Sargan test, regress the IV residuals from part (g) on all instruments. The test statistic is nR^2 , with degrees of freedom equal to the number of surplus instruments.

```
Z_capm_sargan <- data.frame(
  RANK = capm5$RANK,
  POS = capm5$POS
)

sargan_capm <- sargan_test(iv_rank_pos$residuals, Z_capm_sargan, df_overid = 1)
kable_num(sargan_capm, caption = "Sargan test for surplus IV validity", row.names = FALSE)
```

Sargan test for surplus IV validity

statistic	df	p_value
0.5585	1	0.4549

Answer. The Sargan p-value is about 0.4549. At the 5% level, we fail to reject the surplus IV validity condition.